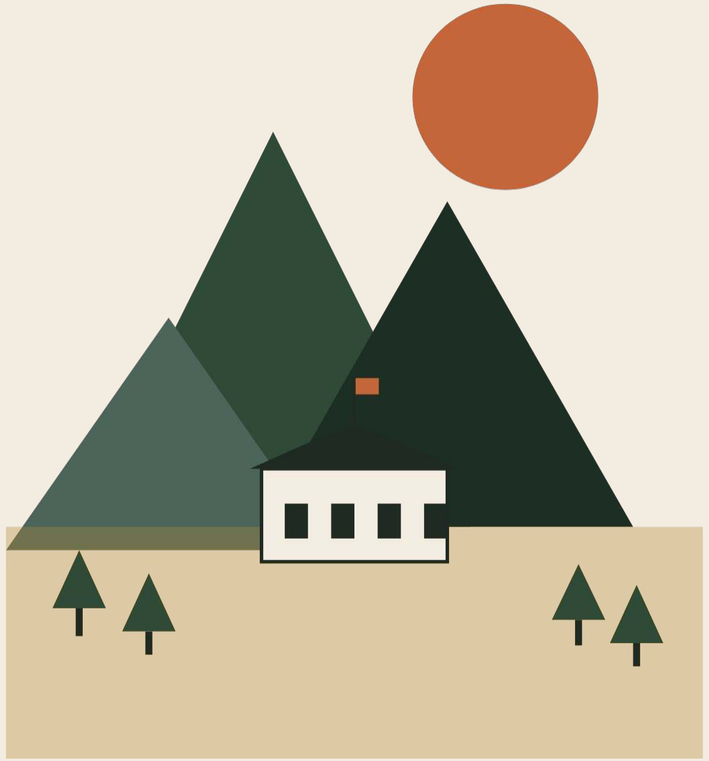
A 25-YEAR INITIATIVE · 二十五年計画

# A modest plan for a small village in the *mountains.*



奈良県御杖村 森林・分散型エネルギー・地域に根ざした小規模データセンター

*"The forest our ancestors planted — powering the village they built."*
*先人が植えた森が、今、村に豊かさをもたらす。*

MITSUE VILLAGE · 奈良県御杖村

# One vacant school building. One *tired forest.* One short list of ideas.

Mitsue is a mountain village in Nara Prefecture. In 2021, the elementary school merged with the junior high and moved to a new building — the old building sits empty. The cedar plantations around it are old and unmanaged. The grid is fragile. None of this is unusual in rural Japan — that's the whole point.

COORDINATES · 座標

**34.4194° N 136.0511° E**

| | |
|---|---|
| POPULATION<br>≈ 1,400 | SCHOOL RELOCATED<br>2020 |
| FOREST COVER<br>≈ 93% | PROJECT HORIZON<br>25 yr |

If a small, careful intervention works here, it should be copy-pasteable into a thousand other villages with the same pattern.

# Four expensive problems that turn out to be *cheaper together.*

四つの構造的圧力、ひとつの農村型解答

01 · ENERGY



Japan targets 100% electrified new car sales by 2035.

In the EVs, PHEVs and FCVs — not pure-EV. Rural grids need distributed generation and charging; extension is slow and expensive.

02 · FOREST



## Cedar is a liability — and a fuel.

Aged sugi plantations: pollen, biodiversity loss, landslide risk. Removing them powers the village; replacing them rebuilds the forest.

03 · STRANDED



## Vacant village buildings just sit there.

Vacant schools and factories cost shrinking municipalities money to maintain. Productive reuse turns them into community anchors.

04 · DIGITAL



## Rural compute is thin.

Broadband and edge-compute capacity lag urban Japan. A small data center closes both gaps at once, on local power.

# Forest. Power. Compute. *One ledger.*

森林再生・分散型再生可能エネルギー・地域に根ざしたデータセンター — 一つの台帳で。

I

## Forest restoration

Aged cedar → native broadleaf, on the ecological clock.

II

## Local energy + EV

Sugi biomass CHP (the trees we remove) leads; solar and EV charging complement. Blackout resilience built in.

III

## Edge data center

A small, community-rooted site — location under research.

# We feed the bears, so the bears *stop visiting.*

Phased replacement of aged sugi (cedar) plantations with native broadleaf — oak, chestnut, konara. Done with local landowners and the Forestry Agency. The 25-year arc is the ecological clock, not the funding cycle.

The sugi we remove doesn't go to waste — it fuels biomass CHP + the data center. That revenue funds higher landowner payouts and the co-op's mechanized equipment.

Broadleaf trees produce acorns and nuts that sustain deer, wild boar, and bear through winter. When the forest feeds them, they stay there — and out of the village vegetable patch.

POLLEN
↓ down

LANDSLIDE RISK
↓ down

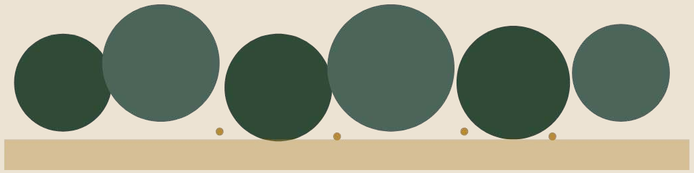
NATIVE CANOPY
↑ up

BEFORE · 杉人工林



Monoculture. Pollen. Quiet ground.

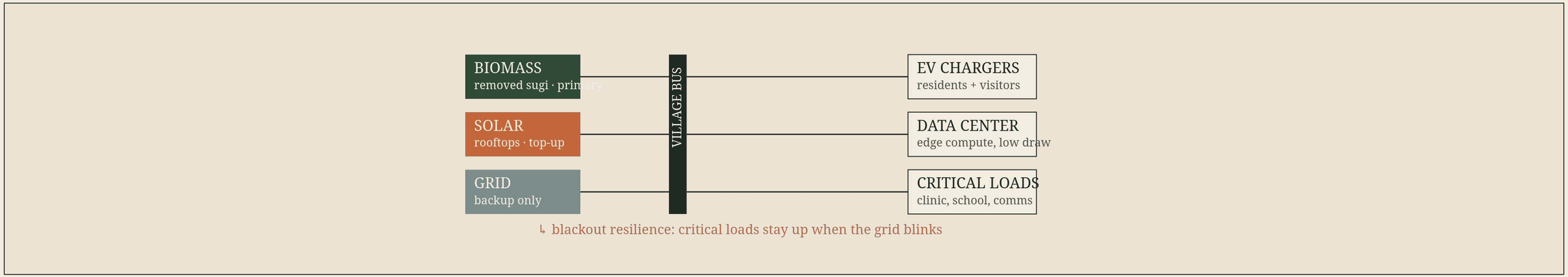
AFTER · 在来広葉樹



Mixed canopy. Acorns. Deer eats here, not your daikon.

# The forest powers the village. *Lights stay on* when the grid blinks.

The sugi we remove during forest restoration fuels a biomass CHP plant that powers a small compute site and EV chargers. We plan to scale in two stages: a ~50 kWe prototype first, then a full-scale CHP of about 1.2 MWe. Solar tops it up; the grid is backup.

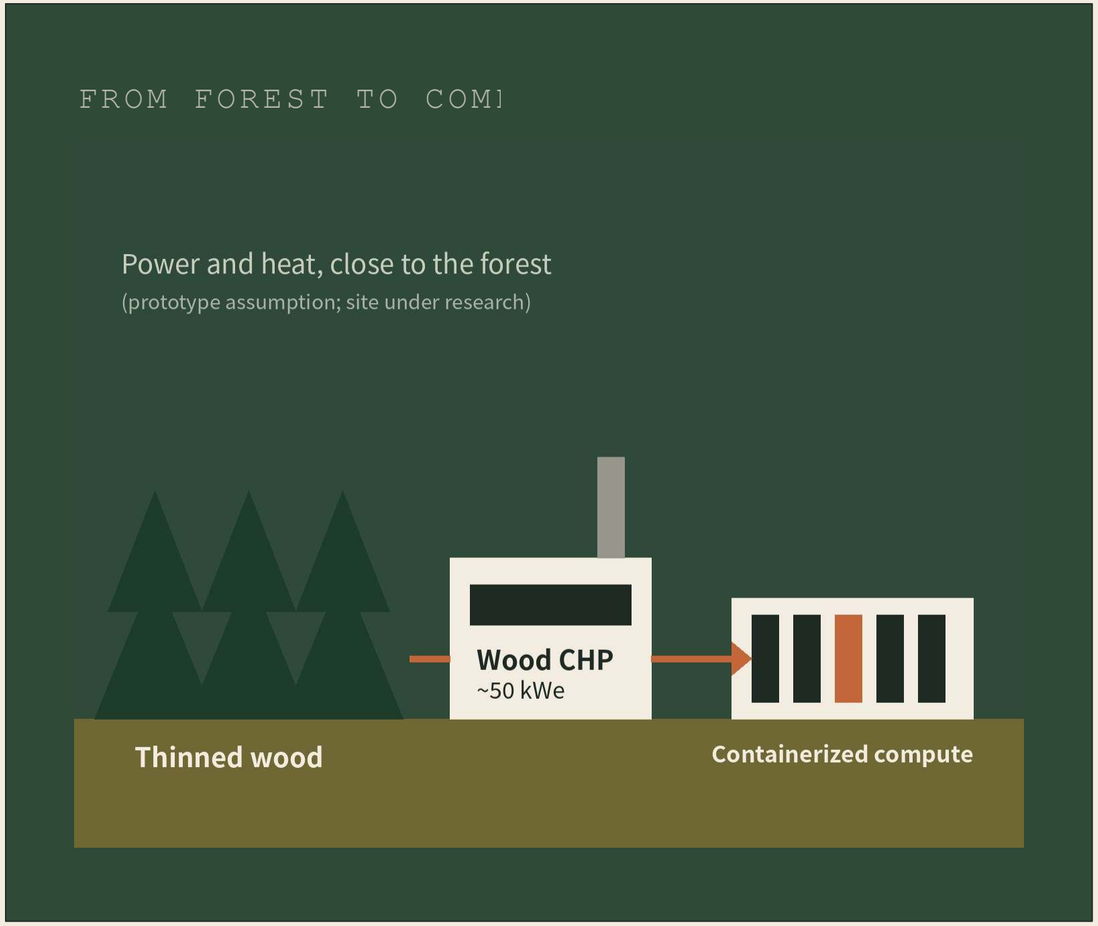


# A data center *powered by the forest.*

"A small data center powered by the forest."

"For the prototype, our current idea is a site belonging to the local forestry cooperative (not confirmed); the location of the full-scale plant is still being researched. It will be a small, energy-efficient edge-compute facility."

Sized for accountability, not hyperscale economics. Small enough for the community to audit and explain what it does. Heat re-use (e.g. drying, heating) is under consideration. Designed to be replicated by other depopulating municipalities — the blueprint is open.




SCALE
Edge

POWER
Wood CHP-led

OPERATION
Community-led

# The win doesn't depend on the data center being a *commercial hit.*

Most benefits accrue to the village whether the compute side breaks even or not. That's the design.

### COMMUNI1

- ↳ Paid forestry work + co-op equipment upgrades
- ↳ EV charging for residents & visitors
- ↳ A new income stream from local resources
- ↳ Blackout-resilient clinic + comms
- ↳ Landowner payouts + a buyer for C/D-grade wood

### ENVIRONME

- ↳ Native broadleaf returns at scale
- ↳ Cedar pollen burden goes down
- ↳ Wildlife pressure on farms drops
- ↳ Landslide risk reduces with age
- ↳ Local woody biomass as the main power source

### BEYOND MITS

- ↳ Methods + data published openly
- ↳ A replicable rural template
- ↳ CC-BY documents, open licences
- ↳ Designed to be copied, not franchised
- ↳ A bridge to small-scale fusion era

PHASE 0 · PRE-FOUNDATION · 準備期

# Quiet on purpose. *No press,* no founder hoodies.

Phase 0 is local trust-building, advisory-board building, and funding-track scouting. Self-funded, ¥0–0.25M. The work that matters right now is meetings, not announcements.

## ADVISORY COMMITMENTS

**Ray Ozzie**
Software pioneer · ex-Microsoft CSA

**Takuo Dome**
Professor, Osaka Univ. · Rep. Director, Inochi Forum

**Henry Takata**
Rep. Director, SynTech Japan · biomass CHP

**Evin Zoet & Yoshiko Zoet-Suzuki**
Co-Representative Directors, Transom

**James Hill**
Osaka Metropolitan Univ. · palaeoecology

**Legal advisors (3)**
Pro bono · entity structure & nonprofit law

## ✓ COMPLETED

- Initial meeting with Vice Mayor of Mitsue (late 2025)
- Mitsue Kanko, local forestry partner, more trees/Tenkawa meetings (May–Aug 2026)
- Founding charter, implementation & EVM plans drafted (BAC ¥220M)
- GX 地域共創 decarbonization subsidy inquiry submitted (Aug 2026)
- Advisory board grown to 11, incl. 3 pro bono legal advisors

## → IN PROGRESS

- Identifying a Japanese co-founder with rural credibility
- Keeping the Village Mayor informed with project documents
- Drafting bylaws for a 一般社団法人 Mitsue-it
- Engaging a 行政書士 (administrative scrivener) in Nara

## ○ NEXT 30 DAYS

- Approach candidate Japanese co-founder
- Continue quiet outreach with village hall
- GX 地域共創 Round 2 application (deadline Nov 30, 2026)
- Prep NAIST seminar (confirmed, Jan 13, 2027)

# Five phases. Four *real* funding gates.

Each gate is a checkpoint. Miss one and we hold and re-pitch — we don't accelerate into an under-resourced phase. The funding stack has five layers, deliberately, so no single source can stall the project early.

PHASE 0 · N
**Pre-Foundation**
M 1–3 · ¥0.25M

PHASE 1
**Foundation**
M 4–9 · ¥5.5M

PHASE 2
**Pilot Design**
M 10–18 · ¥22.25M

PHASE 3
**Pilot Build**
M 19–30 · ¥177M

PHASE 4
**Operate & Scale**
M 31+ · revenue-led

G1 · ¥3–8M　　G2 · ¥30–50M　　G3 · ¥120–290M　　G4 · REVENUE ONLINE

## FIVE-LAYER FUNDING STACK · TARGET ≈ ¥192M · ¥0 SECURED

| Layer | Source | Target | Secured |
|---|---|---|---|
| L1 | Founder / private | ¥6M | ¥0 |
| L2 | Government grants | ¥115M | ¥0 |
| L3 | Foundations | ¥33M | ¥0 |
| L4 | Corporate partners | ¥35M | ¥0 |
| L5 | Operating revenue | ¥3M | ¥0 |
| **Σ** | **Illustrative total** | **¥192M** | **¥0** |

## SIX OPERATING PRINCIPLES

I. Local first

II. Open & transparent

III. Patient, long-term

IV. Replicable

V. Modest in scale

VI. Non-partisan

ALIGNMENT WITH THE VILLAGE'S RENEWABLE ENE

# We don't ask the village to adopt a new idea. We run *the plan it already wrote.*

御杖村再エネ導入最大化計画（2025 年・環境省補助）の官民連携 運営体制として実行します。

i

## One resilient site

The plan targets one renewable + storage + EV resilient site (now zero). Our data center could be that site.

ii

## EV-charging priority

Transport = 46% of village emissions. We deliver the plan's priority EV charging at public facilities.

iii

## Forest J-Credit + 交付金

Biomass CHP supplies the forest-carbon/J-Credit mechanism. Unlocks the village-led 交付金 : 2/3-3/4 subsidy on solar/battery/EV, via the village.



BIOMASS CHP STAYS PRIMARY · SOLAR/BATTERY/EV COMPLEMENTARY · FUNDING FLOWS TH

IF YOUR TIME HORIZON MATCHES

# We're looking for practical know-how, honest critique, and a few *good neighbors.*

WHAT HELPS RIGHT

- ↳ Introductions: Japanese co-founders with rural credibility
- ↳ Advisors: forestry, edge compute, Japanese nonprofit law
- ↳ Technical partners: small data-center operations & power design
- ↳ Honest critique — tell us where the thesis is wrong

SCAN TO VIS

mitsue.it

codeberg.org / YR-Design / mitsue-ai-data-center

info@mitsue.it

PROJECT LEAD NOW

Rob Oudendijk

YR-Design · Safecast · Mitsue, Nara



THANK YOU ありがとうございま